

Technical Notes on the Apparent Hyperfast Motion of Quasars

1. Introduction of the effective refractive index gradient

The starting point of this theory (Spacetime Refraction Model) is to reject the conventional assumption that outer space is a uniform vacuum and to introduce an "effective refractive index gradient" that arises from the interference of external waves (standing waves). When energy radiated from a quasar passes through a spacetime medium with this gradient, it undergoes a local change in phase velocity.

Core concept: The observed passage of time is compressed by the refractive index gradient, resulting in an "amplification of the effective velocity" that transcends the physical velocity of the celestial body itself during the imaging process.

2. Amplification potential R_f Geometric derivation

Amplification potential that determines the maximum amplification capability due to refraction R_f . This is the entire scale of the universe R_{total} . And the reference length of the observation system (galactic orbit) L_{ref} . It is defined as the geometric ratio of .

$$R_f = 2 \cdot \left(\frac{R_{total}}{L_{ref}} \right) \cdot \cot(\theta_{res})$$

Here, L_{ref} This corresponds to the distance traveled during the galaxy's orbital period (approximately 250 million years). θ_{res} This represents the fixed axis (15.0°) of the cosmic resonator. This equation shows that hypervelocity is not an anomaly inherent to celestial bodies, but a physical necessity derived from the geodetic conditions under which our galaxy scans spacetime.

3. Differences between the Standard Model and this Theory

The following comparison table shows the fundamental differences between the standard astrophysical interpretation of quasar hyperfast motion and this theory (WRM).

Comparison items	Standard model (SR/GR)	This theory (WRM V10.0)
Definition of hyperspeed motion	Projection illusions associated with movement in the direction of gaze	Physical amplification due to the refractive index gradient of the spacetime medium
Homogeneity of space	Isotropic, uniform vacuum (Minkowski space)	Anisotropic, non-uniform medium centered on a fixed axis (15.0°)
The role of the observer	Passive inertial frame	Galactic orbital period T_{gal} Dynamic scanning (scanning) is the primary method used.
Interpretation of speed limits	speed of light c is an absolute upper limit	The apparent speed is R_f Physically expandable up to twice its original size.

4. As a dynamic scan coefficient R_f

In the standard model, whether or not we can observe hypervelocity depends on the probability that the jets of celestial objects "happen" to be facing us. However, in this theory, the geodesic timing—specifically, the "phase" at which our galaxy passes through the standing waves of the cosmic resonator—becomes the decisive factor.

This topological interference causes an extreme change in the effective refractive index only at a specific angle (the fixed axis), resulting in effective time compression (i.e., velocity amplification).

5. Geodetic classification of quasar observation data

Based on this theory (WRM V10.0), we reclassified the observational anomalies of major quasars currently reported based on the "scan phase" and "intensity of the refractive index gradient" within the cosmic resonator.

Classification Category	Representative celestial body	Observable physical phenomena	Logical interpretation by V10.0
Type-I: Focused type	3C 273, 3C 120	Stable, ultra-high-speed jet motion exceeding 10c	The current phase of the galaxy perfectly coincides with the fixed axis (15.0°). Effective refractive index ⁿ The compression is maximized.
Type-II: High gradient traversal type	3C 279, PKS 1510-089	The simultaneous occurrence of an explosive gamma-ray flare and faster-than-light travel.	Refractive index gradient ^{∇n} It is scanning an extremely steep region. The time compression rate is fluctuating rapidly in a short period of time.
Type-III: Edge-scanning type	PKS 0405-385, J1819+3845	Intense photonic scintillation lasting several hours	We are currently traversing the "edge" of the resonance peak. Tiny geodetic displacements are being magnified and projected onto the observational data.

6. The "Geometry of Phase" that determines the amplification potential.

The above classification depends not on "what" the object being observed is, but on "where we, the observers, are currently looking from." In the following potential amplification equation, the azimuth deviation($\theta - \theta_{res}$)The closer it gets to zero, the more extreme amplification, like that seen in Type-I, becomes apparent.

Effective Velocity $v_{obs} = v_{true} \cdot \left[1 + \left(R_f \cdot e^{-\sigma(\theta - 15.0^\circ)^2} \right) \right]$

Thus, the "anomalous" phenomena in the listed celestial bodies can be explained uniformly as **snapshots of geodetic focus scans in this theory.**

7. Redefining the reasons for non-observation: Topological filters

The fact that hypervelocity is not observed in most celestial bodies is explained in this theory not by a "spatial absence," but by a **"temporal and phase filter" on the part of the observer.**

- **Geometric mismatch:** When the jet ejection angle is outside the resonant cone of the fixed axis (15.0°), compression due to the refractive index gradient does not occur.
- **Topological unreached:** Galactic orbital period T_{gal} In dynamic scans based on this, our observation system is not currently passing through the "amplified phase" of space where the celestial object is located.
- **Energy threshold:** The initial jet energy required to visualize the refractive effect is insufficient.

8. Future verification issues: Validity of the classification

Points to consider: Whether galaxies like 3C 273, currently classified as Type-I (focused), will exhibit the theoretically predicted "tiny fluctuations in amplification" in response to slight phase changes associated with the galaxy's orbit over observation periods ranging from several decades to several centuries.

9. Conclusion

The "spacetime refraction geodesy model (V10.0)" discussed in this technical note liberates hypervelocity motion from being an inherent property of celestial bodies to an **"imaging phenomenon produced by the relative motion of the medium (spacetime) and the observer."**

The "geodetic stride" of our galaxy as it moves within a cosmic resonator. L_{ref} The perspective that this determines the amplification potential redefines the "observer's position," which the Standard Model overlooked. The validity of the classification based on this theory will ultimately be judged by comparing it with precise long-term observational data in the future.

That concludes the 4-page series.